Supplement: Equal Pixels, Unequal Views

S1. Every registered test and where it is reported

All tests were registered in the pre-registration or one of its amendments before any of their numbers existed. "Main" gives the section of the paper. Fit split means `ntire_val_hard`, on which squish was chosen.

question	outcome	reported
proxy: squish-trained beats crop-trained, 3 seeds	passed, +0.078 to +0.093	Main, Training
full fused models on the hard split	passed, +0.074	Main, Training
each external benchmark	SocialRF pass, CommunityAI confounded, Synthbuster fail	Main, Held-out
CommunityAI counts only if size-matched JPEG gain clears zero	did not clear, confounded	Main, Held-out
aspect-preserving presentations keep 75% of the gain	passed, 93 to 100%	Main (one line), S2
squish helps public detectors	UnivFD at chance, B-Free fails	Main, Held-out
CLIP repeat of the proxy comparison and test-time gain	passed	Main, Training
does squish's frequency content explain the gain?	coverage holds, design confounded	S8
does the gain grow with degradation level?	passed, rise holds	Main, What the gain is made of; S9
field of view or shrink ratio, by regression	not identifiable, no result	S8
is the gain spread across degradation families?	JPEG gains more, mostly headroom	S8
is training field of view narrow?	no, 0.391 to 1.000 after augmentation	Main, Table 1
routed fusion keeps the gain and removes the clean cost	failed	Main, clean cost (one line)
multi-scale training matches squish	squish ahead, gap 73–81% closed	Main, Training
crop baseline handicapped at 224?	no	Main, Training
learned combiner over nine windows matches squish	squish ahead, B ties on small images	Main, Training
does the periphery carry the evidence?	failed, wrong geometry	S6
where is the clean cost on large images?	not located	Main (one line), S7
wording corrections with numbers	no condition	applied in text
does the degradation rise survive degradation after squish?	A 86% survives, B about half	S3
coverage against downscale, NTIRE	both effects	Main, What the gain is made of
native 512 window for branch A	+0.042, mixed with view	Main, Training
coverage against downscale, SocialRF	downscale only	Main, What the gain is made of
downscale by 2: degradation removal or field of view?	field of view	Main, What the gain is made of
degradation removal at stronger downscales	larger at 4x than 2x on both branches, not proof	Main, What the gain is made of
enlargement at equal field of view	+0.011, interval includes zero, unresolved	Main, Training
coverage against downscale on CLIP	field of view, small removal	Main, What the gain is made of
downscale on B-Free	+7.9 points on large degraded SocialRF	Main, Held-out

Not run: a third test read, once planned for routed fusion, was not run by decision. Exploratory, not registered: the DINOv2 test-time comparison on the proxies (S4), the CLIP trained-minus-test-time comparison, and a headroom check on the degradation-family test (S8).

S2. Resolution-shortcut details (moved verbatim)

The gain is not only the geometric distortions. The challenge's validation splits add random crops, aspect crops and downscaling, and the hard validation and test splits also add perspective transforms; its

training split has none of these. These change native size and could carry the effect. Grouping the distorted rows of `ntire_val_hard` by their transforms (Table 1), squish gains in every group. On rows that received only photometric transforms the gain is +9.0 points on A and +8.3 on B, with intervals clear of zero. Those rows have a median short side of 784 pixels; large native images are enough to produce the effect.

distortions	n	short p50	A gain	B gain
photometric only	241	784	+0.090	+0.083
downscale, no crop	740	432	+0.095	+0.065
crop or aspect crop	191	320	+0.059	+0.063
perspective	78	536	+0.146	+0.140

Table 1: Squish gain in robust AUC over the centre window, within groups of distorted `ntire_val_hard` rows, frozen crop-trained branches. All eight 95% intervals exclude zero.

Aspect ratio. Squish stretches non-square images, which Grommelt et al. warn introduces frequency artefacts along one axis. To attribute the gain to field of view rather than to the stretch, we registered two aspect-preserving presentations before scoring them: *short-centre* (short side to the input size, then the centre square) and *letterbox* (long side to the input size, padded to a square). We predicted that each would recover at least 75% of squish's gain on both branches. Both did (Table 2): short-centre kept 93% of the gain on A and 97% on B, letterbox 100% and 94%. Against squish, short-centre is 0.7 points lower in robust AUC on A, with an interval that excludes zero, and indistinguishable on B. Letterbox matches squish in robust AUC but loses 0.4 points of clean AUC on both branches. The stretch is therefore not what drives the gain; showing the whole short side is.

	robust gain over centre		clean vs. squish	
presentation	A	B	A	B
squish	+0.092	+0.073	–	–
short-centre	+0.085	+0.070	0.000	0.000
letterbox	+0.092	+0.069	−0.004*	−0.004*

Table 2: Differences in AUC for the aspect-preserving presentations, with the frozen crop-trained branches on `ntire_val_hard` and a paired bootstrap of 2000 resamples. Every robust gain has a 95% interval that excludes zero. * marks a clean difference whose interval excludes zero.

S3. Degradation after squish

Under the degradation grid of the main text the degradation is applied before squish, so squish's downscale could average it away. We registered a second arm that squishes the clean image first and degrades the squished view. The share of the rise with degradation level that survives is 0.86 [0.83, 0.89] on branch A and 0.55 [0.47, 0.62] on B. B's input is 224 pixels, so squish downscales its images 2.3 times more than A's. The registered rule gives A: coverage, B: unresolved. Degradation on the small squished view is not the same strength as native degradation seen through the enlarged window, so this arm removes the downscale's averaging without equalising what the two arms see. The stronger-downscale test in the main text asks the same question directly.

S4. DINOv2 test-time comparison on the proxies (exploratory, moved verbatim)

The CLIP results led us to run the same test-time comparison on the DINOv2 proxies, which we had not registered. Shown the whole image, the DINOv2 crop-trained proxies gain +0.026 to +0.038 robust AUC,

every interval clear of zero, and the gain again grows with size (+0.075 above 752 pixels, +0.023 below 336). At proxy scale, training under squish adds to the test-time change for both backbones: squish-trained proxies beat the crop-trained ones shown the whole image by +0.048 to +0.057 robust and +0.022 to +0.032 clean with DINOv2, and by +0.038 to +0.048 and +0.025 to +0.040 with CLIP, every interval clear of zero. Test-time squish costs the crop-trained proxies 1 to 4 clean points, which training recovers.

backbone	model	robust	clean
DINOv2	crop-trained	0.7994	0.9255
	crop-trained, squish at test	0.8310	0.9087
	squish-trained	0.8841	0.9357
CLIP	crop-trained	0.7482	0.9655
	crop-trained, squish at test	0.8485	0.9324
	squish-trained	0.8924	0.9645

Table 3: Proxy models (one tower, one epoch on a quarter of the training set) on `ntire_val_hard`, mean of three seeds. The DINOv2 test-time row was not registered.

CLIP mechanism check (registered): the training gain is +0.206 above 752 pixels against +0.091 below 336. On `ntire_val` the DINOv2 proxy robust gain was +0.035 to +0.037.

S5. Earlier method changes (moved verbatim)

Before finding the coverage effect we pre-registered three changes aimed at the degradations, each with a written gate. Distilling degraded features toward clean ones lost about one robust point on the proxy, and under a control that gave both views the same window it still gained nothing (−0.2 points against a gate of +1). Conditioning the classifier on a predicted degradation embedding through FiLM tied the baseline (0.803 against 0.804). Weighting the fusion by predicted degradation severity could not clear its gate of +0.5 points even with an oracle (+0.4 points pooled). These are single-proxy results, but together with the probe they suggest the representation was not the bottleneck.

S6. Masking

We registered that masking the periphery of the squished image would cost more than masking its centre. It failed on both branches: masking a central quarter cost about twice as much as masking an outer rim of equal area (A −0.013 [−0.021, −0.004], B −0.012 [−0.020, −0.003]). A 25%-area rim is a thin border, the emptiest part of a photo, so the test compared the busiest quarter with the emptiest. Unregistered: squish with the centre masked still beats the centre window (A +0.069, B +0.055).

S7. Locating the clean cost

On Synthbuster, branch A changes by at most 0.007 on every generator. Branch B loses in two unrelated places, DALL-E 2 (−0.110, 1024 pixels) and Firefly (−0.040, the whole bucket above 1100 pixels). The registered rule required one location and found none. On logits z-scored against each model's own reals, large fakes gain less than small ones (A −5.1, B −10.7 real-sd), and TPR at 1% FPR on Firefly falls (A −0.062, B −0.259). Synthbuster is near ceiling for A, so a null AUC result says little. The test-split comparison of 0.974 and 0.880 also changes final against averaged weights, training round, and a 518 against 512 run setting.

Moved verbatim from the old main text: Split by native size, the real images are not the cause: under crop their mean score drifts toward "fake" as size grows, and under squish it is flat. On their clean images above 1100 pixels, branch A trained under squish scores 1.000 against 0.989 for crop on `ntire_val` (871 images) and 0.978 against 0.981 on `ntire_val_hard` (295 images).

S8. Superseded and unidentifiable tests

Frequency matching. We squished the image, resized it back to native size, and took the centre window, so the window had squish's frequency content at its own small field of view. It recovered none of the gain (A −0.009, B −0.014 against the centre window), which the registered rule read as coverage. Above 752 pixels the round trip leaves a window that is mostly blur, unlike anything in training, and it also costs clean AUC, so the design is confounded. Superseded by the decomposition in the main text.

Shrink ratio against field of view. A regression of the gain on both could not identify them, because they move together. No result.

Degradation families. JPEG rows gain 2 to 5 times more than others. An exploratory headroom check shows most of that excess is headroom: it falls to 1.2 to 1.6 times once the room left to gain is counted.

Text corrections. Ten wording fixes with numbers, applied in the text.

S9. Degradation grid details

Pooled level-averaged gain of squish over the native presentation, none / light / medium / heavy: A −0.0011 / +0.0011 / +0.0075 / +0.0285, B +0.0043 / +0.0076 / +0.0141 / +0.0271, B-Free −0.1139 / −0.1080 / −0.0884 / −0.0346. The sign change on clean images is resolved for A only, in the 528 to 736 pixel bucket (−0.0013 [−0.0027, −0.0003]). A pre-stated prediction that JPEG would rise most failed, and cannot be tested because the levels are not matched in strength (pixel RMSE at heavy: JPEG 5.7, blur 11.1, noise 23.6). On Synthbuster squish loses at every degradation level.